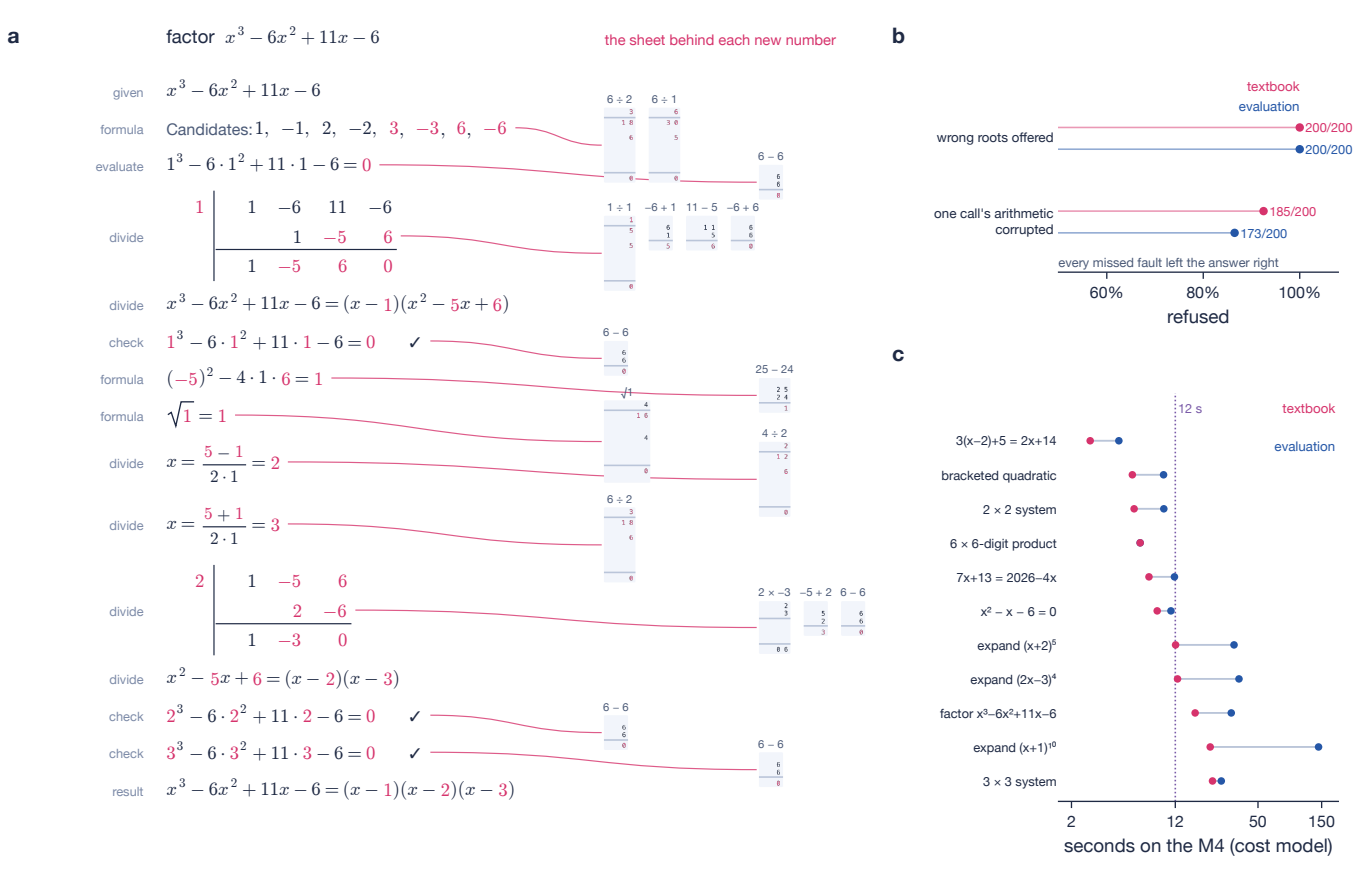

a
factor $x^3 - 6x^2 + 11x - 6$
the sheet behind each new number
given
$x^3 - 6x^2 + 11x - 6$
formula
Candidates: $1,\ -1,\ 2,\ -2,\ 3,\ -3,\ 6,\ -6$
evaluate
$1^3 - 6 \cdot 1^2 + 11 \cdot 1 - 6 = 0$
divide
1 | 1 −6 11 −6
1 −5 6
1 −5 6 0
divide
$x^3 - 6x^2 + 11x - 6 = (x - 1)(x^2 - 5x + 6)$
check
$1^3 - 6 \cdot 1^2 + 11 \cdot 1 - 6 = 0$ ✓
formula
$(-5)^2 - 4 \cdot 1 \cdot 6 = 1$
formula
$\sqrt{1} = 1$
divide
$x = \frac{5 - 1}{2 \cdot 1} = 2$
divide
$x = \frac{5 + 1}{2 \cdot 1} = 3$
divide
2 | 1 −5 6
2 −6
1 −3 0
divide
$x^2 - 5x + 6 = (x - 2)(x - 3)$
check
$2^3 - 6 \cdot 2^2 + 11 \cdot 2 - 6 = 0$ ✓
check
$3^3 - 6 \cdot 3^2 + 11 \cdot 3 - 6 = 0$ ✓
result
$x^3 - 6x^2 + 11x - 6 = (x - 1)(x - 2)(x - 3)$
6 ÷ 2
6 ÷ 1
6 − 6
1 ÷ 1
−6 + 1
11 − 5
−6 + 6
6 − 6
25 − 24
4 ÷ 2
6 ÷ 2
2 × −3
−5 + 2
6 − 6
6 − 6
6 − 6
b
textbook
evaluation
wrong roots offered
200/200
200/200
one call's arithmetic corrupted
185/200
173/200
every missed fault left the answer right
60%
80%
100%
refused
c
12 s
textbook
evaluation
3(x−2)+5 = 2x+14
bracketed quadratic
2 × 2 system
6 × 6-digit product
7x+13 = 2026−4x
x² − x − 6 = 0
expand (x+2)⁵
expand (2x−3)⁴
factor x³−6x²+11x−6
expand (x+1)¹⁰
3 × 3 system
2
12
50
150
seconds on the M4 (cost model)